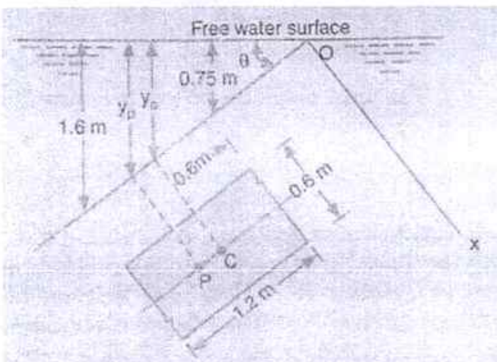
Roll No. | | | | | | | |

End Semester Examination 2024

Name of the Course: B. Tech (Civil Engineering)
Name of the Paper: Fluid Mechanics
Time: 3 Hours

Semester: III
Paper Code: TCE 301
Maximum Marks: 100

Note:
All questions are compulsory
Answer any two sub questions among a, b & c in each main questions
Total marks in each question are twenty
Each question carries 10 marks

Q1		(20 marks)
(a)	Define mass density, specific weight, specific volume and specific gravity.	CO1
(b)	State and prove Pascal's Law and discuss its applications.	
(c)	Two square flat plates with each side 60 cm are spaced 12.50 mm apart. The lower plate is stationary and the upper plate requires a force of 100N to keep it moving with a velocity of 2.50 m/s. the oil film between the plates has the same velocity as that of the plates at the surface of contact. Assuming a linear velocity distribution, determine the dynamic viscosity of oil in poise.	
Q2		(20 marks)
(a)	State and explain the various equilibrium conditions of stability for submerged and floating bodies.	CO2
(b)	Water is flowing through a pipe of 5 cm diameter under a pressure of 29.43 N/cm^2 and with mean velocity of 2 m/s. Find the total energy per unit weight of the water at a cross – section which is 5 m above the datum line.	
(c)	A rectangular plate 0.6 m wide and 1.2 m deep is submerged in an oil of specific gravity 0.80. The maximum and minimum depths of the plate are 1.60 m and 0.75 m respectively from the free surface as shown in figure. Calculate the hydrostatic force on one face of the plate and the depth of center of pressure. 	

Q3		(20 marks)
(a)	Derive an expression for discharge through fully submerged orifice.	CO3
(b)	During an experiment in a laboratory, 0·05 m^3 of water flowing over a right angled notch was collected in one minute. If the head of the sill is 50 mm calculate the co-efficient of discharge of the notch.	
(c)	swimming pool 12 m long and 7 m wide holds water to a depth of 2 m. If the water is discharged through an opening of area 0.2 m^2 at the bottom of the pool, find the time required to empty the tank. Take co-efficient of discharge for the opening as 0.6.	
Q4		(20 marks)
(a)	Describe briefly any two methods of determining the co-efficient of viscosity of a liquid.	CO4
(b)	Prove that the maximum velocity in a circular pipe for viscous flow is equal to two times the average velocity of the flow.	
(c)	of viscosity 8 poise and specific gravity 1.2 is flowing through a circular pipe of diameter 100 mm. The maximum shear stress at the pipe wall is 210 N/m^2. Find: (i) The pressure gradient, (ii) The average velocity, and (iii) Reynolds number of flow.	
Q5		(20 marks)
(a)	What is meant by water hammer? How it can be controlled by Surge Tank.	CO5
(b)	Derive Borda – Carnot equation for head loss due to sudden enlargement in a pipe.	
(c)	Two reservoirs are connected by a pipeline consisting of two pipes, one of 15 cm diameter and length 6 m and the other of diameter 22·5 cm and 16 m length. If the difference of water levels in the two reservoirs is 6 m, calculate the discharge and draw the energy gradient line. Take f = 0·04.	